

Quantum Key Distribution (QKD)

5.1 protocol: BBM92. *arXiv: 1506.08458* for full proof.

this is unrealistic design but gives perfect security

- ① State preparation ✓
- ② Randomisation
- ③ Measurement ✓
- ④ parameter estimation
- ⑤ error correction
- ⑥ privacy Amplification

only ① & ③ is quantum
others just happened in classical stage.
parameters.

$\phi \in \{0,1\}^n, \delta \in (0,1), n=m-k, \text{syndrome: } \{0,1\}^n \mapsto \{0,1\}^r$
 $\pi \in [m], |\pi|=k, k \ll m, \pi \text{ is subset of } m, r \approx n \cdot h(\delta)$
 $m \text{ is \# qubits, } k \text{ is used for } ④, h(\delta) = \delta \log \frac{1}{\delta} + (1-\delta) \log \frac{1}{1-\delta}$

$FPI = \begin{cases} V, & \text{test passes} \\ \perp, & \text{test fails} \end{cases}$

similarly Fec, Fpa



- ② π is random seed
- ③ Measurement:

$\phi_i = 0, M \text{ } A_i/B_i \text{ in } \{|0\rangle, |1\rangle\}$

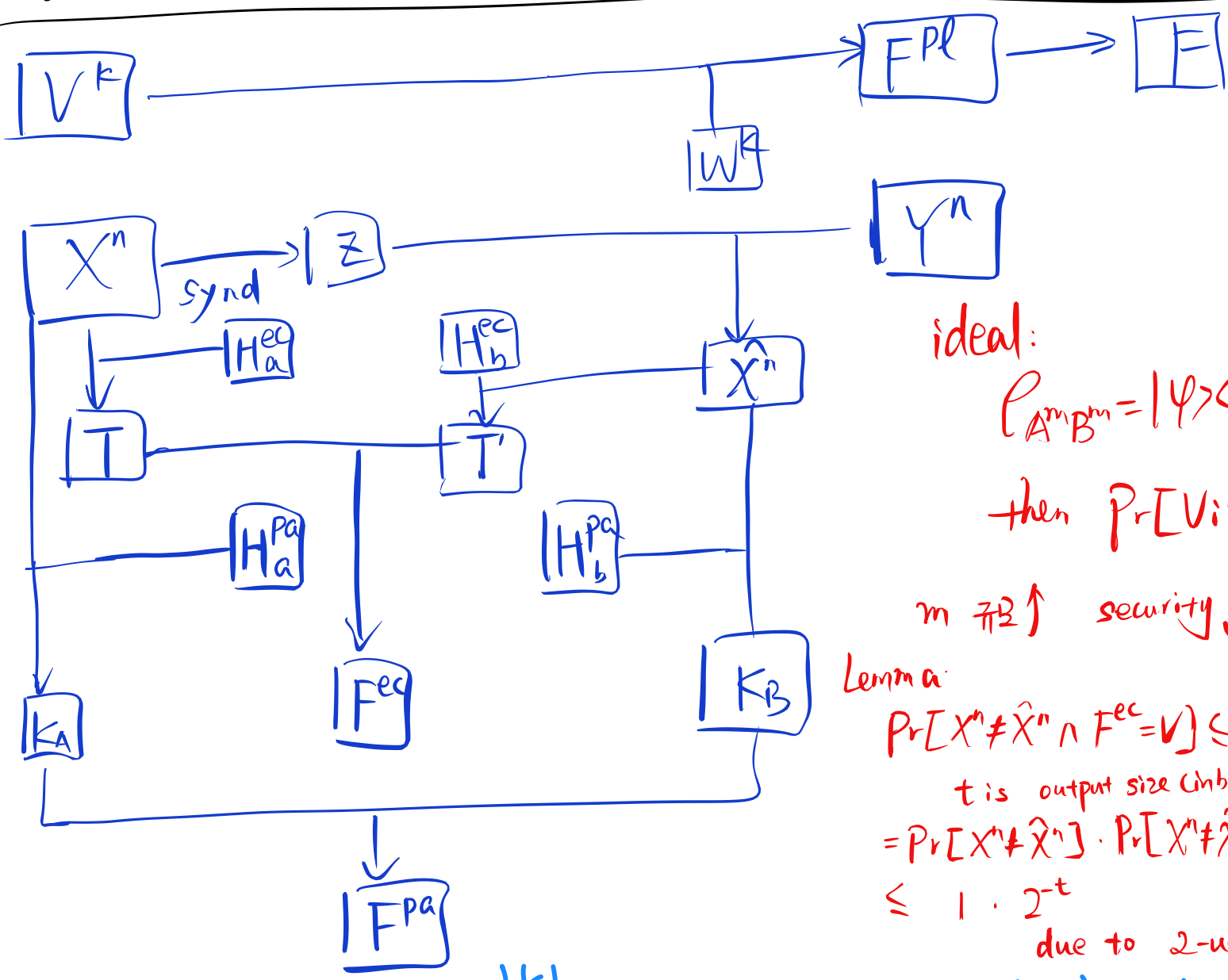
$A^m = \{A_1, A_2, \dots, A_m\}$ m bits string $\phi_i = 1, M \text{ } A_i/B_i \text{ in } \{|+\rangle, |-\rangle\}$

Given Modes:

H_{pa} privacy Amplification, Hash Func.

H_{ec} error correction, Hash Func.

record outcome as $\begin{cases} X_i/Y_i, & i \notin \pi \text{ for } A/B \\ V_i/W_i, & i \in \pi \text{ for } A/B \end{cases}$



ideal:
 $\rho_{A^m B^m} = |\psi\rangle\langle\psi|_{AB}^{\otimes m}$

then $\Pr[V_i \neq W_i] = 0$

$m \uparrow \Rightarrow \text{security} \downarrow \Rightarrow \text{trade-off}$

Lemma:

$$\Pr[X^n \neq \hat{X}^n \wedge Fec = V] \leq 2^{-t}$$

t is output size (nbits) Hash value
 $= \Pr[X^n \neq \hat{X}^n] \cdot \Pr[X^n \neq \hat{X}^n | H^{ec}(X^n) = H^{ec}(\hat{X}^n)]$

$$\leq 1 \cdot 2^{-t}$$

due to 2-universal property

t is key length used in EC.

$$|K| =$$

Assumptions:

- ① Finite dimension quantum mechanism
- ② Sealed laboratories (密封 Lab)
- ③ Random Seeds
- ④ Authenticated classical channel. Bob has to know some msg must comes from Alice, so, requires start from little keys.
- ⑤ deterministic detection. always get outcome.
- ⑥ Commuting Measurement
- ⑦ Complementarity of Alice's Measurement $C = \max_j |\langle \varphi_i | \varphi_j \rangle|^2 = \frac{1}{2}$

①: For any $\dim(H) = d$,

$$P \in \mathcal{L}(H), P \geq 0, \text{Tr}(P) = 1$$

이 proof of security rely on this. otherwise uncontrol/unimagineable events may occur.

②: In case use side channel etc physical attack. Make sure physical isolation

③: ~ ④ ~ ⑤ 시간 위치 정보 ⑥ Z/X 基 上 측정 서로不干渉

⑦: ~ 도청 탐지 eavesdrop.

Goal:

$$\|W_{K_A K_B}^{SCFE} | F=(U,V) - \chi_{K_A K_B} \otimes W_{SCFE} | F=(U,V) \|_{tr} \cdot \Pr[F=(U,V)] \leq \epsilon$$

S. Seeds. C. communication transcripts F. Flags. E. eve's quantum memory

$$\chi_{K_A K_B} = \frac{1}{2^n} \sum_k |k\rangle \langle k|_{K_A} \otimes |k\rangle \langle k|_{K_B}$$

$$\begin{aligned} I = H_{\min}^{\epsilon}(X^n | SCFE) &\geq H_{\min}^{\epsilon}(X^n | E \phi \pi) - r - t - k \\ &\geq H_{\max}^{\epsilon}(X^n | B^n \phi \pi) + n - r - t - k \\ &\geq -H_{\max}^{\epsilon}(X^n | \hat{X}^n) + n - r - t - k \end{aligned}$$

$$\therefore H_{\min}^{\epsilon} + H_{\max}^{\epsilon} \geq n$$

Bob can determine X^n based on his info if he passed before

$$H_{\max}(X^n | \hat{X}^n) \geq (1-h(\delta))n - r - k - t$$

Bounded $h(\delta)n$